Improving Model Fairness with Time-Augmented Bayesian Knowledge Tracing

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ABSTRACT

Modelling student performance is an increasingly popular goal in the learning analytics community. A common method for this task is Bayesian Knowledge Tracing (BKT), which predicts student performance and topic mastery using the student's answer history. While BKT has strong qualities and good empirical performance, like many machine learning approaches it can be prone to bias. In this study we demonstrate an inherent bias in BKT with respect to students' income support levels and gender, using publicly available data. We find that this bias is likely a result of the model's 'slip' parameter disregarding answer speed when deciding if a student has lost mastery status. We propose a new BKT model variation that directly considers answer speed, resulting in a significant fairness increase without sacrificing model performance. We discuss the role of answer speed as a potential cause of BKT model bias, as well as a method to minimise bias in future implementations.

CCS CONCEPTS

• **Applied computing** → **E-learning**; • **Computing methodologies** → **Machine learning algorithms**.

KEYWORDS

knowledge tracing, fairness

ACM Reference Format:
Jake Barrett, Alasdair Day, and Kobi Gal. 2024. Improving Model Fairness with Time-Augmented Bayesian Knowledge Tracing. In *The 14th Learning Analytics and Knowledge Conference (LAK '24), March 18–22, 2024, Kyoto, Japan.* ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3636555.3636849

1 INTRODUCTION

The already fast-growing field of e-learning has expanded significantly during the Covid-19 pandemic. Although students have mostly returned to classrooms, new e-learning challenges abound, such as improving access to education [13], educative efficiency [22], and modeling student performance. A commonly used method for student modeling is knowledge tracing, which can be used for sequencing educational content to students and guiding interventions [2]. Algorithms such as the Bayesian Knowledge Tracing (BKT) model [7, 36] apply machine learning (ML) techniques to map student knowledge with high accuracy. However, such innovations carry a new set of risks and challenges such as the potential for bias and unfairness [43].

While there has been research into the area of fairness and how it interacts with student learning and performance prediction [19, 20], the work done investigating the violations of fairness within a knowledge tracing setting have been lacking. In this study, we consider the varying performance of the BKT model by demographic and social attributes. In particular, we consider both the student's self-reported gender and socioeconomic status as measured by eligibility for income support. Discrimination in AI outcomes along gender lines is well established, both in a wider AI context [29, 38], and the domain of educational AI [4]. Although discrimination based on economic standing is not protected by the Equality Act 2010[1], much work has been done exploring the harm this discrimination can cause [6] [21]. Additionally, there have been calls for socioeconomic status to be added to the list of protected characteristics covered by the Equality Act [27, 28]. Because of this, it is crucial that for any knowledge tracing model to be considered "fair" it must have a consistent prediction accuracy for students with the same mastery level, irrespective of attributes such as income support or gender.

We present three key findings that contribute to existing knowledge in the e-learning applications sphere. Firstly, we identify that existing BKT implementations lead to unfair predictive outcomes for students eligible for income support as well as for male students. Secondly, we posit that this unfairness is caused by an insufficient value of the 'slip' parameter in the standard BKT model, which assumes students who have gained mastery are equally likely to 'slip' and make a careless error regardless of other factors. We show that this is untrue, and identify students' answer speed to be correlated with both answer accuracy and certain demographic variables; this leads to the standard BKT model to underestimate students' knowledge for both students who receive economic assistance and male students. This correlation between a student's answer speed and their performance was identified in previous work in education modelling, which Van Der Linden terms the 'speed-accuracy tradeoff' [41]. Finally, we propose a variation of BKT that rescales

[1] https://www.legislation.gov.uk/ukpga/2010/15/contents

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LAK '24, March 18–22, 2024, Kyoto, Japan
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ACM ISBN 979-8-4007-1618-8/24/03…$15.00
https://doi.org/10.1145/3636555.3636849

the slip parameter by reasoning about answer time. We show this leads to significant improvement in fairness outcomes without requiring any knowledge of a student's demographics and without significantly sacrificing model performance. This method can be used to reduce BKT bias in future implementations.

The empirical investigations in this paper are based on the Eedi competition data set [17]. We used this data to assess the different outcomes in applying a BKT model to student answers, calculating BKT model fairness using Equality of Odds and Equality of Opportunity metrics. For comparisons, we use a baseline model of N consecutive correct responses (N-CCR) for varying values of N, and therefore varying baseline strictness thresholds. Given that a student attains 'true' mastery in the N-CCR models, we demonstrate that the BKT model is likely to under-predict mastery status if a student is male or if a student is in receipt of income support, with a larger difference for the latter group. We then demonstrate that the relationship between answer speed and bias is causal by explicitly including answer speed in a time-augmented BKT model.

The rest of this work is structured as follows. In Section 2 we consider the current art in educational interventions, focusing on knowledge tracing and fairness metrics. We introduce the background to our model in Section 3, with a detailed look at BKT in Section 3.1, our proposed baseline "true mastery" metric in Section 3.2, arguing that N-CCR provides a lower bound on mastery attainment, and fairness metrics based on previous works in Section 3.3, to obtain metrics for model fairness evaluation. We then develop our model in Section 4, introducing our proposed extension to BKT in Section 4.1. We present our computational results in Section 5, fitting our model to the Eedi competition data set [17]. Finally, we discuss our work's consequences for equitable e-learning and future implementations in Section 6.

2 RELATED WORK

We present two key strands of literature that relate to this work: previous work on knowledge tracing, focusing on BKT, and how to measure the fairness of algorithmic interventions in the educational AI sphere.

2.1 Knowledge Tracing

One of the simplest forms of knowledge tracing, which we leverage in this paper as a baseline, is defining mastery through a student having attained N consecutive correct responses (N-CCR) for varying strictness thresholds N [23]. Despite its simplicity, this implementation is known to perform well in comparison to more sophisticated metrics [23, 32] and, because of its simplicity, the scope for unintentionally introducing model bias is low. This approach is widely used in existing e-learning assessments, such as the ASSISTments platform [12] and Khan Academy [11].

Beyond N-CCR, there are multiple parameterised approaches available, of which we focus on BKT [7]. This approach models student performance as a Markov process with parameterized estimates for initial knowledge, knowledge gain, knowledge slip, and the probability that the student is guessing. We focus on BKT due to both its intuitive parameter interpretations and its extensive use in the field: both Pelánek [31] and Abdelrahman et al. [1] provide wide-ranging surveys of modern BKT techniques and implementations. Deep knowledge tracing is also well-studied in the literature, with strong accuracy performance across tasks including modelling student knowledge [33] and providing question recommendations [2]. For the purposes of exploring fairness in our study, and particularly identifying the root cause of any bias, we prioritise BKT over deeper approaches due to higher model and parameter interpretability [25].

We extend the BKT model by adding a variable for answer speed, following previous BKT extensions in the literature. Qiu et al. [34] consider two extensions to BKT, labelled KT-Slip and KT-Forget, which respectively consider a rustiness effect on the first question of the day and a time-based 'forget' parameter introduced to deal with new sessions. However, this work is concerned primarily with model performance rather than fairness. Tschiatschek et al. [39] investigate the effects of a modified BKT model tuned individually to each students, labelled Bayesian Bayesian Knowledge Tracing (B^2KT). In this model the parameters are continually updated throughout the run-time, rather than one-shot optimisation, to try and represent each student more accurately. They demonstrate model benefits compared to regular BKT, but concede that this type of research may not be enough to result in truly equitable tutoring systems. Gal et al. [8] also explore an individualised implementation of BKT, integrating students' repeated answer attempts to provide individual question recommendations. In utilising an individualised slip parameter based on answer speed, our approach extends these three works.

2.2 Measuring Fairness

Hardt et al. [15] provide a central point of reference for fairness in AI, exploring demographic parity (the tightest fairness constraint, requiring the predictor to be independent of a protected attribute), Equal Odds (requiring that the positive predictor is independent of the protected attribute given the observed outcome) and Equal Opportunity (requiring that the positive predictor is independent of the protected attribute given that the observed outcome is positive). Of these, Equal Odds provides empirically strong incentives for fairness, in settings such as credit risk [15] and child welfare [6].

In the specific application of educational AI, Madaio et al. [26] explore fairness issues in a comprehensive study, focusing on historical imbalances and highlighting the importance of a robust end-to-end system including inclusive design and usability. Holstein and Doroudi [16] consider how long-established problems in wider AI (biased data sets, human-in-the-loop issues, platform access disparity) can cause algorithms to perpetuate existing unfairness. Several other works have formalised fairness in educational technologies using machine learning: Hutt et al. [19] model participation dropout predictions by comparing area under the ROC curve (ROC-AUC) for male and female model predictions, looking at courses with different gender balances. Predictive bias arises from AI models prioritising accuracy for the majority group in each course. Sha et al. [37] take a similar approach for automatic education forum post classifiers, finding that prediction accuracy was generally fair across gender and biased towards first-language English speakers. Hutt et al. [14] use this approach for predictions of on-time graduation, investigating predictive accuracy by student

ethnicity and socioeconomic status. Jiang and Pardos [20] apply Equal Odds and Equal Opportunity to assess student grade predictions, and other works apply this lens to identify at-risk students and college success [18, 44].

Doroudi and Brunskill [9] show that both N-CCR and BKT algorithms perform differently with different learner types, providing too many easy questions to above-average learners before assigning mastery, thus wasting student goodwill and resources. Their results indicate the BKT model underestimates certain learners, which we also observe in a bias setting. We directly extend their work in applying a formal notion of fairness to the knowledge tracing task and demonstrating its inherent bias, replicating their results on a new data set and for student demographics (rather than behaviours), and demonstrating the source of this bias and how to correct for it.

Finally, a key philosophical consideration of any fairness metric is discussed by Dwork et al. [10]. In this work, the authors consider the fairness-privacy relationship through the problem of balancing equitable outcomes with awareness of individuals' demographic properties, where knowledge of the latter can lead to discriminatory practices. They thus emphasise the importance of metrics that limit the use of direct demographic information, which is a key pillar of our approach.

3 KNOWLEDGE TRACING TOOLS

We present three key background subjects across both knowledge tracing and fairness. First, we introduce the BKT framework in its raw form (3.1). We then introduce our baseline metric of mastery acquisition, N-CCR, in Section 3.2. We introduce our fairness metrics used to assess these models in Section 3.3.

3.1 Bayesian Knowledge Tracing

Bayesian knowledge tracing (BKT) models students' acquisition of mastery using a temporal probabilistic approach [7]. The key difficulty in predicting mastery in BKT is that it is a latent variable, meaning it cannot be directly observed from data and rather must be inferred from other variables. BKT computationally derives a (per-student, per-topic) confidence value $P(L_{i,j,k}) \in [0, 1]$ corresponding to the likelihood that student i has achieved mastery of a given topic k by their j^{th} answer attempt. Chronologically ordering the student's attempts at questions in a topic, the model predicts whether the student has answered the jth question correctly and then updates its predicted probability that the student has mastered the chosen skill based on their answer history and their probability of guessing of making a mistake. This thus yields a probability of mastery of topic k at each question response j.

The core BKT equations are characterised by the following parameters:

- $P(C_{i,j,k})$: the probability that student i answers their j^{th} (chronological) question in topic k correctly
- $P(G_k)$ (the guess parameter)[2]: the probability of a correct guess in topic k if the student does not know the skill
- $P(S_k)$ (the slip parameter): the probability of an incorrect answer in topic k despite the student knowing the skill
- $P(T_k)$ (the learn parameter): the probability of the student learning topic k at any given step

$P(G_k), P(S_k), P(T_k)$ and the prior probability of mastery $P(C_{0,k})$ are initialised at pre-defined values and then optimised during training using a method such as expectation-maximisation (EM). After training they remain constant for future model usage. The dynamics of the BKT model are given in Equations 1 and 2.

$$P(C_{i,j,k}) = P(G_k)(1 - P(L_{i,j,k})) + (1 - P(S_k))P(L_{i,j,k}) \quad (1)$$

$$P(L_{i,j,k}) = P(L_{i,j,k} - 1) + P(T)(1 - P(L_{i,j,k} - 1)) \quad (2)$$

To demonstrate the predictions made by this model, consider an example student who answers 15 questions in the following pattern: [0,1,0,1,1,1,1,1,0,1,0,0,1,0,0] (with a 1 indicating a correct answer, and a 0 indicating an incorrect answer). When this student's performance is passed into an off-the-shelf BKT implementation, it produces mastery predictions as demonstrated in Figure 1. These predictions indicate a near-total confidence in mastery after the first 8-10 answers, which falls away as the student rounds off their performance with mainly incorrect answers. We leverage this example student throughout the rest of the paper, considering what this performance means in the context of N-CCR and our time-augmented model.

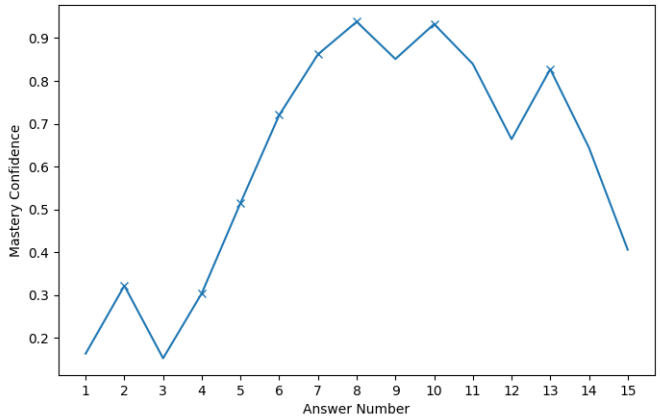

Figure 1: Example mastery predictions produced by BKT model; correct answers are marked with a cross

3.2 N Consecutive Correct Responses

To evaluate the accuracy of the trained BKT model, and therefore fairness outcomes, we require a 'true' baseline value of student mastery. Quantifying mastery in a truly objective manner is very difficult, and we thus employ subjective model-free metrics following both real-world implementations [11, 12] and approaches in the literature [9] to inspect results and fairness with varying strictness thresholds for mastery. We calculate the maximum number N of consecutive correct responses on a per-topic basis that each student has achieved. Higher values serve as a lower bound for mastery; as a student achieves an increasing number of consecutive correct

[2]Note that the guess, slip, and learn parameter are all time invariant in the classic BKT model: it is assumed that the probability of learning a skill, correctly guessing, or accidentally slipping in a question answered quickly is the same as for a question answered slowly.

responses, our confidence that mastery has been achieved increases. In this work, we utilise two thresholds for N-CCR: a weak threshold of $N = 5$, and a strong threshold of $N = 10$.

We stress the importance of correctly interpreting our use of the N-CCR metric. Particularly for high N values, the metric represents a lower bound for mastery: if a student has consistently answered correctly for a prolonged period, they have at least attained mastery by their N^{th} consecutive correct response, and likely attained it at a much earlier stage. While this will naturally lead to a higher 'false negative' predictive rate, in our fairness considerations via Equality of Opportunity we are interested in equitable predictive outcomes for those who have attained mastery. Therefore, although it may seem arbitrary to not assign mastery to those with $9 - CCR$ under our strong threshold, this increases confidence that our fairness investigations are only considering students who have truly achieved mastery.

If we consider the example student used in Section 3.1, the highest consecutive number of correct responses they achieved was 5. Therefore, at a cut-off of $N = 5$, this student is judged to have achieved mastery, and a more lenient reader may be satisfied by this definition. However, the student followed these 5 correct responses with 5/7 incorrect responses; at a cut-off of $N = 10$, mastery in this topic is not judged to have been attained. Whether the student has truly attained mastery based on this performance requires a value judgement, but it is clear that the $N = 10$ threshold is a strong threshold for mastery confidence compared to the weaker $N = 5$.

The N-CCR metric is not without flaws, as like BKT it will be affected by factors such as lucky guesses, forgetfulness, or test-taking stress. Nonetheless, it provides a customizable strictness as our lower bound for true mastery attainment, and its simplicity and interpretability make it a compelling baseline model. Additionally, as shown by Kelly et al. [23] it provides accuracy on par or above that offered by more complex measures.

3.3 Fairness Metrics

In the AI context bias occurs when two "similar" inputs are treated dis-similarly: if relevant factors are consistent between inputs A and B, A and B should have the same output, even if they differ on some factor x which is irrelevant to the prediction being made. Should A and B have a different output, this is identified as a bias on factor x. For this paper, two metrics of fairness will be considered, as defined by Hardt et al. [15]: Equality of Opportunity and Equality of Odds.

Equality of Opportunity is satisfied when students who have truly mastered a skill are predicted to have mastered it irrespective of any demographic variables. Equality of Odds is a stricter measure, which is satisfied when mastery predictions are consistent irrespective of both the student's true mastery and any demographic values.

Formally, in a knowledge tracing context, Equality of Opportunity requires that students who have mastered a given skill ($L_{i,j,k} = 1$) to have the same predicted value for mastery in the model, $\hat{L_{i,j,k}}$, regardless of their (binary) demographic variable D:

$$P(\hat{L_{i,j,k}} = 1 \mid D = 1, L_{i,j,k} = 1) = P(\hat{L_{i,j,k}} = 1 \mid D = 0, L_{i,j,k} = 1) \tag{3}$$

For Equality of Odds, the requirements are made stricter so that regardless of the "true" mastery status $L_{i,j,k}$, students of different values for demographic D have equal access to a correct prediction $\hat{L_{i,j,k}}$:

$$P(\hat{L_{i,j,k}} = 1 \mid D = 1, L_{i,j,k} = l) = P(\hat{L_{i,j,k}} = 1 \mid D = 0, L_{i,j,k} = l), \quad l \in \{0, 1\} \tag{4}$$

To understand the demographic bias present, an odds ratio (OR) is calculated across the demographic value in question. To compute Equal Opportunity in this way, only cases where the student's "true" value of mastery is positive ($l = 1$) were examined, but for Equal Odds all cases regardless of "true" mastery $l \in \{0, 1\}$ are examined.

$$OR(D, l) = \frac{P(\hat{L_{i,j,k}} = 1 \mid D = 1, L_{i,j,k} = l)}{P(\hat{L_{i,j,k}} = 1 \mid D = 0, L_{i,j,k} = l)} \tag{5}$$

For the demographic value D, an odds ratio above 1 shows that the model is overly assigning mastery to students of demographic group $D = 1$ irrespective of actual performance (and vice-versa a value below 1 would show favouritism to $D = 0$). Should the ratio be sufficiently far from 1 then this would be evidence that the model is biased over demographic variable D. Likewise, a model without bias would have an odds ratio of approximately 1.

We chose this fairness approach for two reasons. First due to its consistent use in work applying fairness to BKT models [15]. Second, under Equal Odds and Opportunity, false positive and true positive rates must be the same across groups. However, these could potentially be quite significantly different for "good" students under demographic parity, but different in the other direction for students with lower achievement rates, which contravenes the spirit of our fairness investigations.

4 METHODOLOGY

In this Section, we introduce our contribution to extend BKT to include answer speed. We then define our model setting in Section 4.1. Based on the time-accuracy tradeoff in the literature around education [41], and correlations between available time and demographic variables [5, 29], we hypothesise that including answer time in the BKT model will lead to improved fairness outcomes.

4.1 Time-augmented BKT

The BKT slip parameter controls the probability that the student has answered a question incorrectly despite having mastered the skill. In a regular BKT model this parameter is a static value across each topic k. However, there is compelling evidence from the literature (Section 2) and the Eedi data that a time-invariant slip parameter can unintentionally compound unfairness effects in algorithmic mastery assessment approaches: students qualifying for income support are more prone to rushing through questions, perhaps due to lower time availability from a more hectic home life that results in less ideal studying conditions and/or less time available to spend on homework [5]. This pattern is also observed for male students [30]. These factors could negatively impact their answers causing a misrepresentation of their actual mastery. We therefore introduce a dynamic time-augmented slip parameter to the BKT model.

Software to fit the BKT model to data, such as the pyBKT software we use in this work [3], recursively moves through all of a student's answers for a given topic, using an expectation-maximisation (EM)

approach to continually update the posterior probability that the student has mastered the skill based on each answer's correctness. Defining $O_{i,j,k}$ as the observation of a student i's answer to topic k at time j, the algorithm recursively solves the equation pair 6 and 7. To augment this structure, we have added a flexible student-specific slip scaling coefficient denoted σ_i for student i.

$$P(L_{i,j,k} \mid O_{i,j,k} = 1) = 1 - \frac{(1 - P(T_k))(1 - P(L_{i,j-1,k} \mid O_{i,j-1,k}))P(G_k)}{P(G_k) + (1 - \sigma_i \cdot P(S_k) - P(G_k))P(L_{i,j-1,k} \mid O_{i,j-1,k})} \quad (6)$$

$$P(L_{i,j,k} \mid O_{i,j,k} = 0) = 1 - \frac{(1 - P(T_k))(1 - P(L_{i,j-1,k} \mid O_{i,j-1,k}))(1 - P(G_k))}{1 - P(G_k) - (1 - \sigma_i \cdot P(S_k) - P(G_k))P(L_{i,j-1,k} \mid O_{i,j-1,k})} \quad (7)$$

In standard BKT implementations, there is no scaling factor for the slip parameter: $\sigma_i = 1 \ \forall i$. In our augmented model, the slip parameter is scaled up or down depending on agent i's answer speed t. In an ideal world, this would scale as a smooth function of speed, but because the Eedi data set aggregates response times to the nearest minute we introduce a distinct speed coefficient value $\sigma_i = L$ for slow ($t \leq 1$), $\sigma_i = M$ for medium ($1 < t \leq 2$), and $\sigma_i = F$ for fast ($t \geq 2$) answer speeds. These parameters are tuned in our model application, and cut-offs are chosen due to data availability: only 4.46% of all answers have an answer time greater than 3 minutes but less than 10 minutes.

To demonstrate the effect of this changed model, let us consider again the example student described in Section 3.1. Suppose they were assigned 15 homework questions and having performed well on the first 10 questions - achieving 5 consecutive correct answers and only making three mistakes - ran out of time and had to answer the remaining 5 questions in just a few minutes, resulting in rushed answers and only 1 correct response. Most people watching this play out would likely be happy that the student had demonstrated their mastery regardless of the final 5 questions, but the BKT model would continue lowering the predicted probability of mastery over these final questions, resulting in a lower final prediction.

The augmented model will provide different mastery predictions depending on answer speed. As such, we can graph the predictions made on our example student if they speed up and rush the last five questions, alongside the predictions made if they answer at a consistent speed throughout: the effects of an augmented model with slip scaling values of $S = 0.5$, $M = 1.5$, and $F = 2.5$ is shown in Figure 2.

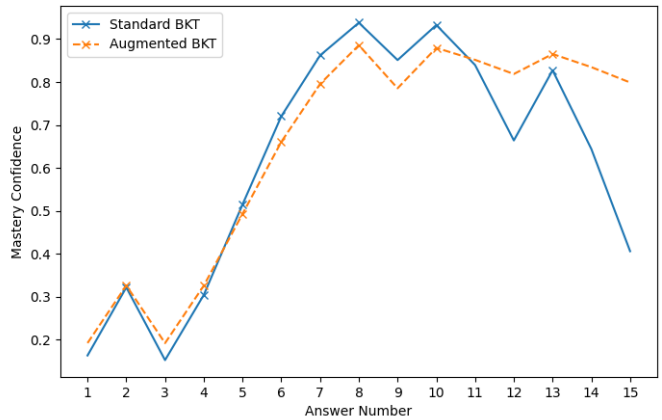

Figure 2: Example predictions produced by standard BKT and the time-augmented model, for a student who rushed the last five answers. BKT model predictions are shown on the y axis, and the (time-ordered) answer attempt number is shown on the x axis.

As shown in Figure 2, the modified model follows the default BKT model very closely for the first 10 answers. For the final 5 answers, each question is answered in under a minute, representing a scenario where a student has run out of time in their homework; the slip parameter is therefore scaled upwards by a factor $F = 2.5$. While the predictions still drop over the course of these five questions, the drop is far less substantial than the default model. This results in a much higher, and more accurate given the hypothetical situation, final prediction of mastery.

We construct the BKT model on an 80% training set, with values split on unique user IDs as the model requires all of any one student's answers to make predictions. This split provides 3,934 training users with 1,104,209 answers, and 984 test users with 274,016 answers. All randomization was performed with fixed seeds for reproducibility, and code is available at our GitHub.[3]

For standard BKT, we implemented a straightforward open-source implementation pyBKT (a Python implementation of BKT) to carry out modelling [3]. For the time-augmented model, we tested a range of possible cut-off values for the slow/medium/fast slip parameters, before optimising the model over a grid search of parameter values using an EM algorithm.

We simultaneously calculate N-CCR values for our weak and strong thresholds, at every time step for each user. Equality of Odds can be computed by splitting the predictions by the income support eligibility demographic and then taking a ratio of their average values. For Equality of Opportunity, the same process is completed, but first the student/subject pairs are filtered such that they only include the entries that satisfy that required baseline mastery. We conduct 5-fold cross-validation for each model setup to test the robustness of our approach.

Finally, for defining answer time for the augmented slip parameter, any answer that took over 10 minutes was viewed as the start

[3]https://github.com/jake126/BKT_fairness

of a new session and therefore not included in the calculations as there is no way to tell how long the student spent on it.

4.1.1 Eedi data set. For this project the model was trained and evaluated using the Eedi data set [17]. This data set was collected via an online learning tool featuring multiple-choice maths questions. This tool is primarily deployed in schools and is used by children from ages 7 — 18. Students' answers are collected along with metadata regarding the questions, the subjects in the questions, the students, and the student's answers. More specifically, this study utilises the Eedi data for 12 — 13-year-olds.

The data is provided in the form of multiple files each pertaining to one category of information. The extensive quantity of data made it suitable for all desired analysis but also resulted in redundant data that was not useful. Accordingly, experiments were primarily performed using a master dataframe composed of the following fields:

- Order: Acts as the index for the dataframe
- QuestionID: Uniquely identifies each question in the data set
- UserId: Uniquely identifies each user in the data set
- AnswerId: Uniquely identifies each answer in the data set
- IsCorrect: A binary value, indicating whether each answer is correct or incorrect
- Name: The name of the subject that that particular question tests
- SubjectId: Uniquely identifies each subject in the data set
- DateAnswered: The date and time of the answer submission aggregated to minutes
- Gender: An integer value indicating the student's gender
- PremiumPupil: A binary value, indicating whether the student is eligible for financial aid e.g. free school meals.

To assess fairness outcomes, we examined two variables: Gender and PremiumPupil. In the Eedi data, there are 1,880 female students and 1,884 male students, with 12 respondents in the category 'other': due to low sample size for this category, we focus on male and female comparisons. There are 733 students in receipt of income support ('Premium pupils'), while 1,808 are not in receipt of income support. Note that these total counts are unequal: we exclude from our fairness analysis students who have unknown values for these variables.

There are reasons to suspect bias for both of these variables. Historic gender bias in AI applications is well-established [29, 38], and has been demonstrated in education-specific settings [4]. For income support status, students receiving income support are more likely to rush their homework, due to correlated demographic factors [5]; this result has also been demonstrated for gender, with female students consistently spending more time on homework [30].

5 RESULTS

In this Section we show three results that support our three hypotheses that standard BKT implementations have unfair outcomes (5.1), that this is caused by an answer-time invariant slip parameter (5.2), and that adjusting this parameter can lead to improved fairness outcomes without significantly sacrificing model performance (5.3).

5.1 Results without Time Augmentation

	5 CCR	10 CCR
True positive, Premium pupils	0.659	0.842
True positive, Non-Premium pupils	0.745	0.924
$OR(D_I, 1)$	1.131	1.098
True positive, Female pupils	0.748	0.919
True positive, Male pupils	0.709	0.897
$OR(D_G, 1)$	1.055	1.025

Table 1: Odds Ratio and Predictions of the classic BKT model considering student performance for those who have mastered skills according to differing N-CCR baselines (3 d.p.)

In Table 1, we report the Odds Ratio and predicted probability that a student has attained mastery in a subject according to the BKT model, given that the student has attained mastery according to the N-CCR metric for both $N = 5$ and $N = 10$. This gives the true positive rate $P(\hat{L_{i,j,k}} = 1 \mid D = d, L_{i,j,k} = 1)$ for $d \in \{0, 1\}$ and $D \in \{D_I, D_G\}$. For both levels of model strictness, BKT under-predicts mastery for Premium pupils. Given that a student attains N-CCR, they are respectively 13.1% and 9.8% more likely to be predicted mastery if they are not in receipt of income support for $N = 5$ and $N = 10$. BKT also under-predicts mastery for male pupils, albeit to a lower extent: a student with 'true' mastery is 5.5% more likely to be predicted mastery if female at $N = 5$, and 2.5% more likely at $N = 10$.

Although we focus on Equal Opportunity, the stronger fairness measure of Equal Odds is also not satisfied. For students who have not attained $5 - CCR$, students without income support are 38.8% more likely to be predicted mastery ($OR(D_I, 0) = 1.388$), while for $10 - CCR$ the discrepancy is over 40% higher ($OR(D_I, 0) = 1.432$). Thus, even when neither student has actually attained mastery, Premium pupils are significantly less likely to receive a mastery prediction. Female students without 'true' mastery are also 15.6% more likely to receive a mastery prediction than male students at $N = 5$ ($OR(D_G, 0) = 1.156$) and 12.4% more likely at $N = 10$ ($OR(D_G, 0) = 1.124$).

Thus this model fails to satisfy both equality of odds and equality of opportunity, and is not a fair model when analysed over the income support eligibility demographic.

5.2 Differing Slip Performance by Demographic

We hypothesise that unfair outcomes are driven by an insufficiently defined slip parameter in BKT. To demonstrate this correlation, we here show differing student performance. Table 2 shows how student performance varies for different answer speeds given income support status. Two metrics demonstrate the key trends: the proportion of all answers with answer speed t, and the corresponding proportion of correct answers with answer speed t. We report the proportionate difference between each rate across the demographic subgroups of interest. Values above 1 in the first (third) row indicate that non-Premium (female) pupils are giving a higher proportion of all their answers with the corresponding speed than Premium (male) pupils, while values above 1 in the second (fourth) row indicate that

non-Premium (female) pupils are giving a higher proportion of all their correct answers with the corresponding speed than Premium (male) pupils.

		$t \leq 1$	$1 < t \leq 2$	$t \geq 2$
Income Support	All answers	0.922	1.485	1.566
	Correct answers	0.926	1.329	1.378
Gender	All answers	0.951	1.271	1.289
	Correct answers	0.953	1.196	1.228

Table 2: Different student performance for different answer speeds in minutes. Values represent differences in the proportions of responses at each answer speed, while the second row represents differences in correct answer proportions at each answer speed.

Trends in answer rates are the same for both Income Support and Gender. Under 1 minute is the only time period for which both Premium and male pupils answer a higher proportion of their questions than non-Premium pupils. Aggregating all answer lengths above 1 minute, the proportion of non-Premium pupils answering in that time is around 50% higher than that of Premium pupils. The figure for male pupils is 37%, which is a significant gap. Both of these trends represent significant behavioural disparities.

The second row of Table 2 represents the proportions of all correct answers with answer length t, with values representing the proportionate difference across demographic variables. The trend here is very similar to the proportions of all answers: both Premium and male pupils give a slightly higher proportion of their correct answers in the fastest category, while both non-Premium and female pupils' correct answers are more likely to occur at slower speeds.

This reinforces the idea that both Premium and male pupils have less time in general to commit to these exercises, affecting both their answer behaviours in general and the pattern of their correct answers.

5.3 Results with Time Augmentation

We now demonstrate that the relationship between an inadequate slip parameter and model unfairness is causal by adjusting this parameter and showing improved fairness outcomes.

Based on exploratory analysis of student performance at differing speeds, the scaling values of $F = 2.5$, $M = 1.5$, and $S = 0.5$ appear natural and give intuitive interpretations. For example, if the base slip parameter is 0.1 and the model is making predictions on a question that was answered in under a minute, then the slip parameter would be increased to 0.25, meaning the model views this answer as more likely to be the result of a careless mistake than others, and therefore calculates a more favourable prediction of mastery at that step. Alternatively, if the same model is predicting for a question answered in 4 minutes, then the slip parameter would be decreased to 0.05, meaning the model views this answer as less likely to be the result of a careless mistake than others, and therefore calculates a less favourable prediction of mastery at that step. To calculate the baseline slip parameter, we used the average value across all subjects for each of the other model parameters in the augmented model.

The fairness of the augmented model is examined in comparison to the original model in Table 3. We focus on Equality of Opportunity, and present the odds ratios across both gender and income support status for pupils who have mastered skills according to different thresholds are compared across the two models. The odds ratios of the augmented model are lower for both thresholds of N-CCR, providing causal evidence that bias in the original BKT model is due — in at least some capacity — to the disregarding of answer speed. We can confirm our results using the cross-validation comparisons. For income support, using a Wald test to test for statistically significant differences in test statistics, the odds ratio is different for 5CCR and 10CCR at the 5% significance level. For gender comparisons, the odds ratio improves for all five folds and CCR levels, but odds ratio differences were not sufficient to reject the null hypothesis of equal means.

		5 CCR	10 CCR
Income Support	Classic BKT	1.131	1.098
	Augmented BKT	1.063	1.057
Gender	Classic BKT	1.055	1.025
	Augmented BKT	1.023	1.001

Table 3: Odds ratios $OR(D, 1)$ of both models considering student performance for those who have mastered skills according to differing N-CCR baselines (3 d.p.)

Finally, to demonstrate that this improvement in fairness is not at the expense of significant model accuracy [42], we compare the performance of the new model with pyBKT, we report the mean squared error (MSE) for each model in Table 4. The performance of the two models is comparable, with the new model providing slightly more generous predictions overall as shown by a higher error on a stricter bound of mastery. Additionally, the accuracy of the new model could almost certainly be improved upon with access to more granular answer time data. This improvement in fairness with a relatively low difference in model accuracy is an intuitive result, as we are feeding more information into the model, indicating that a higher overall accuracy should be achievable without jeopardising other metrics of interest. This outcome is also consistent with other empirical observations of negligible accuracy reductions when fairness is prioritised in the public policy domain [35]. These results are statistically significant at the 5% significance level across 5-fold cross-validation, with an increase in performance at 3CCR, no change at 5CCR (which is commonly used in the field), and a slight fall in performance at 10CCR.

	5 CCR	10 CCR
Classic BKT	0.142	0.195
Augmented BKT	0.147	0.242

Table 4: Mean Squared Error (MSE) of both models when calculated across our N-CCR baselines (3 d.p.)

6 DISCUSSION

This study investigated the fairness of BKT model predictions when evaluated on the fields of both income support and gender. We found that the BKT model is inherently biased when evaluated with the metrics of equal odds and equal opportunity, and therefore is not a "fair" model. Specifically, for any two students A and B who have attained mastery, where A is eligible for income support and B is not, the model is 9-13% more likely to predict mastery for B than for A, while if A is male and B is female, the model is 2.5-5.5% more likely to predict mastery for B than A. A large part of the reason for this is that both male students and students eligible for income support tend to answer the questions more quickly, a factor that the model fails to account for. Results are statistically significant for income support across 5-fold cross-validation, and although the null hypothesis of improved performance is not rejected at the 5% level for gender, this is largely due to a conservative overestimate of sample variance. More realistic variance measures that take the data volumes into account lead to a rejection of the null hypothesis; we were keen to establish a 'worst case' baseline for model improvements.

This finding demonstrates a significant problem with the BKT model, as both male students [40] and students eligible for income support [5] already have lower average educational performance, differences that would only be reinforced by unfair models. As such, it is clear that work must be done to incorporate more fairness considerations into BKT before it is appropriate for widespread use in student learning tools. One such way to do this — as demonstrated in this study — is changing the model to actively consider answer speeds when making its predictions. This solution has shown the potential for a strong improvement in fairness while maintaining similar accuracy to the original model. Additionally, our results have been achieved with a limited data set, meaning they represent a lower bound on possible fairness improvements. This means that should similar improvements hold across other data sets, it is likely that BKT models could provide a practical, useful, and most importantly fair tool to be used within the setting of AI-assisted education.

The model discrepancies are significantly larger for income support than gender, and there are many possible ways to justify a stronger connection between income support eligibility and more rushed answers in this setting. It seems most likely that students eligible for income support — who on average will come from lower socioeconomic backgrounds — will have less time to spend studying and fewer tools to aid with their studying such as laptops, tablets, etc. Additionally, these students can have a higher likelihood of additional responsibilities and disruptions, such as being young carers [5, 24], which will further detract from their time available for homework. Male students are also less likely to either have the means or willingness to prioritise homework and extra-curricular educational activities [30], due to social-cultural factors and differing leisure behaviours outside school hours.

We draw attention to the relevance of this result in the context of fairness through awareness, as discussed by Dwork et al. [10] In other fairness in ML approaches, there is a trade-off between model fairness and differential privacy: using demographic information as inputs can improve fairness outcomes, but gives rise to ethical concerns. In our results, improved fairness outcomes are achieved both without a significant reduction in model accuracy and without requiring demographic input into the model.

6.1 Future Work

The next logical step in this area would be to further explore the effect that dynamic parameters can have on BKT model performance and fairness. The results provided in this report are promising, but leave room for both the accuracy and the fairness of this model to be improved further with more fine-tuning. One potential approach would be to train a model on data that has more precise timing information so that a direct relationship between answer speed and slip chance may be calculated. We prioritised the Eedi data set due to its rich demographic information, but now that improved fairness has been demonstrated it would be interesting to explore how far it is possible to improve performance with granular time data. Additionally, there is much room for experimentation with different optimization techniques regarding the trainable model parameters and how they interact with the slip parameter now that is no longer a static value.

As well as optimizing the model on the current data set, this approach can extend to different data sets. In particular, it is important to test whether the fairness improvements for gender and income status come at the expense of other demographics not available in the data set. Ensuring that fairness improvements scale over different demographics, and particularly intersectional demographics, is a key next step. We note that our output improved fairness outcomes across both demographics considered, which lends promise to this extending to other demographic groups in different data sets when tested in future work.

Beyond optimizing current approaches, there is also large capacity for extending this approach to applications that derive from BKT. For example, it would be valuable to apply this fairness lens and methodolgy to deep knowledge tracing (DKT), as deep learning models are criticised for prioritising performance over transparency and fairness. Should these suggested improvements be implemented successfully, our approach can represent a real and lasting improvement to fairness within the field of AI-assisted education.

ACKNOWLEDGMENTS

We thank Eedi for their publicly available student data from the 2021 Neurips Educational Challenge https://eedi.com/projects/neurips-education-challenge.

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Received -; revised -; accepted -